Gabriel Agostini

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EDUCATION

Cornell University, Cornell Tech — May 2027 (expected)
PhD in Information Science with minor in Urban Planning
Advisors: Emma Pierson & Nikhil Garg

Cornell University, Cornell Tech — 2024
Masters of Science in Information Science

Columbia University, Columbia College — 2022
Bachelor of Arts in Urban Studies | GPA: 4.06/4.00

Columbia University, School of Engineering and Applied Sciences — 2021
Bachelor of Science in Applied Mathematics | GPA: 4.17/4.00

Budapest Semesters in Mathematics — 2020

ACADEMIC PAPERS

*equal contribution

Gabriel Agostini, Rachel Young, Maria Fitzpatrick, Nikhil Garg, Emma Pierson. "Inferring fine-grained migration patterns across the United States." *Nature Communications 17 (*1265), 2026.

Talia Kaufmann*, **Gabriel Agostini***, Trivik Verma, Dan O'Brien. "Inequality in mobility patterns: reconciling access and travel in American cities." *19th International Conference on Computational Urban Planning and Urban Management (CUPUM)*, London, United Kingdom. June 19, 2025.

Gabriel Agostini, Emma Pierson, Nikhil Garg. "A Bayesian Spatial Model to Correct Under-Reporting in Urban Crowdsourcing." *38th Annual AAAI Conference on Artificial Intelligence (AAAI24)*. **Oral presentation**. 2024.

Rajiv Movva*, Sidhika Balachandar*, Kenny Peng*, **Gabriel Agostini***, Nikhil Garg, Emma Pierson. "Topics, Authors, and Institutions in Large Language Model Research: Trends from a Survey of 17K arXiv Papers." *Annual Conference of the North American Chapter of the Association for Computational Linguistics* (NAACL). 2024

Gabriel Agostini, Juliana Gonçalves, Trivik Verma. "Identifying Urban Morphology from Street Networks with Graphlet Analysis." *30th Annual Geographical Information Science Research UK (GISRUK)*, Liverpool, United Kingdom. April 1, 2022.

Gabriel Agostini, Yushan Zhang, Debra Laefer. "A Category-Theory Approach to Construction Ontologies in Subsurface Mass Transit." *International Archives of Photogrammetry, Remote Sensing, and Spatial Information Sciences*, XLVI-4/W4-2021, 125–130, 2021.

IN PREPARATION

Gabriel Agostini, Nikhil Garg, Emma Pierson. "Modeling Biker Preferences with Large Scale E-Bike GPS Data". To appear in *EAAMO 2026* (non-archival).

Gabriel Agostini*, Zhixuan Qi*, Katelyn Liu, Shreya Sriram, Esther Li, Emma Pierson, Nikhil Garg. "Quantifying Spatiotemporal Gaps in Accessibility with Fine-Grained, Open-Source Amenity Datasets"

Erica Chiang, Katelyn Liu, Anum Ahmad, Zhi Liu, **Gabriel Agostini**, Tyler Gibson, Uma Bhandaram, Nikhil Garg. "The Forest from the Trees: Lessons on Public-Sector Deployment from an Academic Collaboration with NYC Department of Parks and Recreation"

PUBLIC SECTOR EXPERIENCE

New York City Department of Transportation Cycling & Micromobility 2025 - Present
Fellow, College Aide | Supervisors*:* ***Carl Sundstrom****,* ***Mark Seaman***

- Built an automated pipeline to simplify NYC street network and incorporate bike facilities
- Developed a general map-matching algorithm suitable for large-scale GPS ping data.

Stanford & San Francisco Unified School District Partnership 2021
Stanford University Data Science for Social Good | Supervisor: ***Prof. Irene Lo***

- Designed algorithms to ensure equal access to education for underserved students in San Francisco.
- Employed spatial analysis techniques to identify geographical proxies for educational barriers.

RESEARCH EXPERIENCE

Cornell Tech 2023 - 2026
Cornell University | Supervisors: ***Prof. Emma Pierson, Prof. Nikhil Garg***

- Pursue independent research on computational social science topics
- Supervised 8 graduate and undergraduate students on individual thesis and internship projects

Centre for Urban Science and Policy 2020 - 2021
TU Delft University | Supervisor: ***Prof. Trivik Verma***

- Collect, process, and study street networks with reproducible Python code
- Developed geometrical and graph-theoretical classifications of cities according to morphology

Urban Modeling Group 2020 - 2021
Center for Urban Science and Progress, NYU | Supervisor: ***Prof. Debra Laefer***

- Characterize subway stations via an ontology of objects and their relationships to improve BIM systems
- Leveraged concepts from categorical theory (ologs, functors) into civil engineering

Mathematics Research Experience for Undergraduates May - August 2019
Math Department, Columbia University | Supervisor: ***Prof. Kyle Hayden***

- Prove open conjectures in knot theory with analytical and computational tools
- Wrote Python scripts to perform topological operations on abstract objects

TEACHING EXPERIENCE

Cornell Tech, Teaching Assistant Fall 2022, Fall 2023, Spring 2024
Courses: Applied Data Science, Fairness in Machine Learning Algorithms

Columbia University Mathematics Department, Teaching Assistant Spring 2019 - Spring 2022
Courses: Multivariable Calculus, Linear Algebra, Ordinary Differential Equations

Fênix Pré-Vestibulares, Mathematics Instructor Summer 2018

COLLABORATIONS IN THE ARTS

Backslash at Cornell Tech, Associate producer for BADWIZARD 2025 - Present
Artist collaborator: Tianyi Sun

PiTech Arts and Culture Fellowship 2025 - 2026
Artist collaborators: Lincoln Center Collider Fellows

AWARDS & HONORS

IPUMS Spatial Student Research Award, IPUMS — April 2026

International Geographic Information Funds Student Paper Award, AAG-IGIF — March 2026

Phi Beta Kappa, Columbia College — April 2022

Summa cum Laude, Columbia College — April 2022

Summa cum Laude, Columbia University School of Engineering and Applied Sciences — May 2021

SERVICE & LEADERSHIP

Urban Data EAAMO-Bridges Working Group, Organizer — 2024 - Present
Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)

EAAMO Conference, Proceedings Chair — 2025
Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)

Cornell Information Science Graduate Application Review Program, Mentor — 2023, 2024, 2025
Cornell Information Science Student Council

INVITED TALKS

Penn State University, EEPI Seminar — April 2026

New York University, Urban Lab: Immigration and Housing in Queens — March 2026

Vision Zero Initiative, NYC Research on the Road — November 2025

Cornell Tech, Urban Systems — September 2025

Barnard College Center for Computational Science, CSC Workshop — September 2025

Cornell Tech, Applied Data Science — October 2024, September 2025

Cornell Tech, Urban Data Science — November 2024

FELLOWSHIPS

PiTech Arts and Culture PhD Fellowship — Fall 2025 - Spring 2026
PiTech: Public Interest Technology Initiative at Cornell Tech

Siegel-PiTech PhD Impact Fellowship — Summer 2025
PiTech: Public Interest Technology Initiative at Cornell Tech

Stanford Data Science for Social Good Fellowship — Summer 2021
Stanford Data Science Institute

Tandon Undergraduate Summer Research Program — Summer 2020
New York University Tandon School of Engineering

RELEVANT COURSEWORK

Analysis & Topology	Real Analysis (A), Measure Theory (A+), Complex Analysis (A+), Functional Analysis (A) ODEs (A+), PDEs (A+), Dynamical Systems (A+), Topology (A+)
Algebra	Linear Algebra (A+), Abstract Algebra: Group Theory (A), Abstract Algebra II (A+)
Statistics	Probability Theory (A), Stochastic Processes (A+), Time Series Analysis (A+) Statistical Inference (A+), Multivariate Statistics (A), Sample Surveys (A)
Network Science	Graph Theory (A+), Spectral Theory of Graphs and Groups (A+), Structure of Complex Networks (A+), Analysis of Networks and Crowds (A)
Machine Learning	Machine Learning (A), Geometric Data Analysis (A), Bayesian Optimization (A+)
Urban Studies	Global Urbanism (A), Urban Elsewheres (A), History of the City in Latin America (A) Colonial Cities of the Americas (A), Postwar Fiction and Architecture (A)

SKILLS

Programming	Python (matplotlib, seaborn, scipy, pysal, networkx, shapely, geopandas, osmnx, sklearn), R, Stan, SQL
Technology	Git, Microsoft Excel, ArcGIS, QGIS
Languages	Brazilian Portuguese (native), Italian (advanced), Russian (basic)